

An AI Based Irrigation and Weather Forecasting System utilizing LoRaWAN and Cloud Computing Technologies

Ala' Khalifeh
School of Electrical and Information
Technology
German Jordanian University
Amman, Jordan
ala.khalifeh@gju.edu.jo

Abdullah Al-Qammaz
School of Electrical and Information
Technology
German Jordanian University
Amman, Jordan
Abdellah.alqammaz@gju.edu.jo

Khalid A. Darabkh
Dept. of Computer Engineering
The University of Jordan
Amman, Jordan
k.darabkeh@ju.edu.jo

Laith Abualigah
Dep. of Computer Engineering
Amman Arab University
Amman, Jordan
Aligah.2020@gmail.com

Ahmad M. Khasawneh
Dep. of Software Engineering and
Mobile Computing
Amman Arab University
Amman, Jordan
a.khasawneh@aau.edu.jo

Zinon Zinonos
Dep. of Computer Sciences
Neapolis University Pafos
Pafos, Cyprus
zinon.zinonos@nup.ac.cy

Abstract— Artificial Intelligence (AI) has been flourishing recently as a viable solution for many applications and scenarios, including smart irrigation and weather forecasting systems. In these systems, it is crucial to have an accurate prediction for the weather and soil conditions to optimize the irrigation process such that the minimal amount of water is provided. In this paper, a smart irrigation system utilizing Artificial Intelligence (AI) and Long Range Wide Area Network (LoRaWAN) communication link is proposed. The system is composed of sensors that are used to measure soil moisture, atmosphere temperature, and humidity. This information is sent via LoRaWAN communication link to a remote center that gathers, analyzes the captured information, quantifies the appropriate amount of water for irrigation, and then sends the decision back to the irrigation system. Furthermore, the collected information will be stored in the cloud for wider accessibility. This paper describes the technical implementation of the smart irrigation system and focuses on the weather forecasting process, which is performed using the Wind Driven Optimization - Least Square Support Vector Machine (WDO-LS-SVM) algorithm. The obtained results show a better performance when compared to the LS-SVM, which verifies the effectiveness of jointly utilizing the WDO with the LS-SVM.

Keywords— *Wind Driven Optimization, WDO, Least Square Support Vector Machine*

I. INTRODUCTION

Smart irrigation systems are widely used recently, due to their effectiveness in optimizing the water resources needed for the irrigation process. One of the most popular irrigation method, surface irrigation, is also the least effective. By wetting areas where no plants will gain wastes a high percentage of water.

On the other hand, localized irrigation would more safely and reliably use water, eliminating under-irrigation and over-irrigation. Farmers, however, feed more water than is needed to prevent loss of production under irrigation. Consequently,

efficiency is not only threatened, but water is still lost. Technology should then be established and implemented to detect the water level demanded by the plantation and flow the water in the appropriate quantity to areas where and when needed.

The new invention of the Internet of Things (IoT) in precision agriculture helps us enhance the overall management of agriculture [1]. According to its highly interoperable, modular, pervasive, and transparent nature, IoT enabled systems are excellent fit for precision agriculture [2]. Many IoT-derived innovations offer many advantages, including minimizing the risk of vendor lock-in, machinery proliferation, and improved systems for sensing/automation [3]. We developed our smart irrigation system, inspired by the above advantages and potential of the IoT used in precision agriculture, despite the lack of a complete, efficient, well-established, and standard solution.

In many agricultural cropping systems in semiarid and arid countries, irrigation is an integral activity, and sufficient water applications and management are key issues. Water is typically relatively consistently applied by self-propelled center pivot and linear-move irrigation systems; however, there are significant differences in soil properties and water supply across most fields [4-6]. In these situations, the potential to apply spatially and temporally variable conditions in site-specific irrigation management will maximize application efficiencies, decrease environmental effects, and even boost yields.

There are currently several obstacles that still prohibit the mainstream use of IoT for precision irrigation from being resolved [7]. Firstly, the production of IoT-based smart applications, such as agricultural irrigation, is not entirely automated. Second, to simplify part of the process and incorporate numerous technologies such as IoT, big data analytics, cloud computing, and fog computing to launch pilot apps for smart water management, sophisticated IoT

development systems are still lacking. Thirdly, it needs appropriate specifications and knowledge templates to combine heterogeneous and specialized sensors.

In the literature, several optimization methods have been successfully used in solving various problems. These algorithms have different inspirations and procedures. One of the well-known optimization methods that can be used to solve many optimization problems is the Wind Driven Optimization Algorithm [8]. This algorithm has been successfully used to solve several optimization problems, including electromagnetics, image segmentation, load frequency control, estimation of solar photovoltaic parameters, 0-1 knapsack problem, global optimization, and feature selection and engineering problems.

This paper complements our previous work [9-13] that aimed at designing and implementing an energy efficient wireless sensor network that can be used for various applications such as environmental monitoring, border security, pollution detection, etc. However, our previous work focused mainly on optimizing the network connectivity without considering how data are analyzed and how decisions can be drawn. Further, our previous work proposed network architecture that utilizes low power low distance communication protocols such as ZigBee. In this paper, we leverage our contribution by analyzing the captured data and proposing a prediction model that could be used in several applications such as smart irrigation. In addition, we propose a network architecture that utilizes the long-distance low power communication protocol such as the Low Range Wide Area Network (LoRaWAN), which is extremely useful especially in open, far and remote areas, which is the case for smart agriculture and irrigation systems.

The aim of this paper is to present a holistic view of the proposed smart irrigation and agricultural system with an emphasis on the weather forecasting process, which is a vital stage that is often used in smart irrigation [14]. The proposed weather forecasting is achieved using the Wind-Driven Algorithm and Least Square Support Vector Machine (WDO-LS-SVM), which we shows a superior performance over the LS-SVM prediction algorithm that is widely used in weather forecasting [15,16].

The rest of this paper is organized as follows: Section 2 presents the most related works in this domain. Section 3 presents the proposed smart irrigation system for weather forecasting. Section 4 discusses the proposed WDO-LS-SVM algorithm. Section 5 demonstrates the methodology used for performing the weather forecast. The results are presented and discussed in Section 6. Finally, Section 7 concludes the paper and proposes future work.

II. LITERATURE REVIEW

This section reviews some well-known related work, which has been published previously. The authors of [17] presented a smart IoT irrigation system designed specifically for urban areas whereby remote IoT devices do not have immediate internet access or power grid, and where long distances and many barriers make wireless communication complicated. It proposed a LoRaWAN-based architecture that offered long

distance and low power consumption communications to solve such problems. In particular, the system is composed of IoT nodes that gather sensor data and forward them to local fog computing nodes or a remote cloud to specify an irrigation schedule that takes into account parameters such as the weather forecast or the humidity detected by the sensing nodes.

Agricultural irrigation is the world's largest freshwater consumer, calling for the extensive use of technology to improve the use of water, reduce power consumption and increase the quality of crops. Although the Internet of Things as well as other related technologies are the obvious option for smart water management systems, with the implementation of on-site pilots, their acceptability is still to be demonstrated in real settings. IoT-based systems for application creation should also be sufficiently standardized to be adapted to various crops, climates, and countries. The SWAMP project designed IoT-based smart water management techniques and practices in precision irrigation and has piloted them in Italy, Spain, and Brazil. Authors in [18] presented the SWAMP view, design, pilots, and the scenario-based creation methods employed in the project in this work.

The development of wireless sensor network (WSN) technology provides a sustainable solution to greenhouse Precision Agriculture (PA). It enables to use agricultural resources and management tools efficiently and tracks various parameters to achieve better yield and performance outcomes. WSN allows the use of Low-Power Wide-Area Networks (LPWANs), a wireless system with limited energy consumption to transfer data over long distances. Despite its low data rate and because of its low implementation and maintenance costs, LoRaWAN is among the most popular LPWAN technologies. Greenhouses are vulnerable to various forms of intrusion and diversification, requiring an improved design scheme for WSN. In [19], authors considered the viable challenges for PA in greenhouses and suggested the successive steps that are necessary for the successful deployment and facilitation of WSN.

Smart freshwater management for precise irrigation in agriculture is important to increase crop yield and reduce costs, thus contributing to the sustainability of the environment. The extensive use of technology provides a way of supplying plants with the exact amount of needed water. For smart water management applications, the Internet of Things is the obvious alternative, even when the integration of various techniques needed to make it works seamlessly in operation is still not completely achieved. For instance, the SWAMP project mentioned earlier creates an IoT-based smart water management system based on four pilots in Brazil and Europe for precise irrigation in agriculture with a hands-on approach. This paper [20] presented the design, platform, and device implementations of the SWAMP that illustrated the platform's comparability, and provided a performance evaluation of FIWARE modules used in the platform, as scalability is a major concern for IoT applications. Findings demonstrated that it could provide the SWAMP pilots with adequate performance but needs specially designed setups and re-engineering of some modules to provide greater scalability using less computing cost.

In recent years, IoT has also found an important place in the agricultural and food production system in addition to its implementation in smart city applications. A revolutionary, power-efficient, and highly scalable IoT agricultural system is presented in [21]. This model is based on the LoRaWAN network for the transmission of long-range and low power consumption from sensor nodes to cloud services. For analytics purposes, the proposed cloud service system is scalable and uses data streams. It has seen some initial findings for grape farms in the experimental of this work.

III. SYSTEM DESIGN AND ARCHITECTURE

In this section, a brief description of the proposed smart irrigation system is presented. Figure 1 depicts the proposed system, which consists of the following main components: LoRaWAN sensor nodes, LoRaWAN gateway, Chirpstack framework, and finally a database and cloud integration. In what follows, a brief description of each component is provided.

A. LoRaWAN sensor node

It is the sensor node responsible for collecting data via different sensors. The node consists of a microcontroller, Waspnote in our case [22], sensor board, and set of sensors to measure the humidity, temperature, pressure. The node also can be connected to an anemometer to measure the wind speed and any other relevant sensors. These sensors are connected to the microcontroller using a sensor board. Furthermore, the microcontroller is connected with a wireless LoRa module transceiver that enables it to communicate with the LoRaWAN gateway.

B. LoRaWAN Gateway

Once the sensor node captures the data, it sends it to LoRaWAN gateway, sometimes named as the Packets' Forwarder. In the proposed system, we used the MultiTech gateway [23], which has 16 different wireless communication channels, thus enabling several LoRa nodes to communicate with the gateway simultaneously. The main role of the gateway is to collect the data and forward it to the gateway bridge for further processing and analysis.

C. ChirpStack framework

It is an open-source framework that provides several services needed for LoRaWAN operation [24]. These services

include but are not limited to: interfacing with the LoRaWAN gateway, providing the users with a web interface for network management, Application Programming Interfaces (APIs) for application developers, and cloud interfacing functionality and database connectivity. Furthermore, the collected data are used by the end application to perform related application functionalities such as weather prediction and irrigation decisions.

D. Database and cloud integration

In order to have broad access to the captured data, a cloud network can be integrated into the system such as ThingSpeak [25], ThingsBoard [26], and Grafana [27]. The later one is adopted in the proposed framework. Once data are processed and saved, a smart irrigation decision is made and sent back to the irrigation system via the LoRa communication link, where another node at the water irrigation pump side receives the decision and controls the amount of water needed for irrigation.

IV. WIND DRIVEN ALGORITHM LEAST SQUARE SUPPORT VECTOR MACHINE (WDO-LS-SVM) ALGORITHM

Least Square Support Vector Machine (LS-SVM) is the least square version of SVM using in data analysis and pattern recognition, and it is known that is easy to train and provide less computational cost [28]. In LS-SVM, a solution of the Quadratic Programming (QP) problem of the SVM is found by solving a set of linear equations. On another side, LS-SVM is considered as a kernel-based learning classifier, and it is used for classification and regression tasks. LS-SVM showed a superior performance in most of the studies within different fields. However, the key important effect of its performance is the tuning of its parameters, including smoothing parameter σ and regularization parameter γ of the Radial Basis Function (RBF) kernel, as this kernel is shown to be one of the most powerful and efficient kernels used in LS-SVM. Thus, these two parameters need to be ideally tuned.

Therefore, several studies depend on grid search method to tune these parameters [29, 30]. Meanwhile, other studies utilized an optimization algorithm intending to optimally tune the parameters that should provide a better performance of the classifier in a specific regression or classification task [31, 32]. Hence, Wind Driven Optimization (WDO) algorithm is proposed to be used for tuning the LS-SVM parameters in this study. As a result, this study provides a new regression scheme named WDO-LS-SVM.

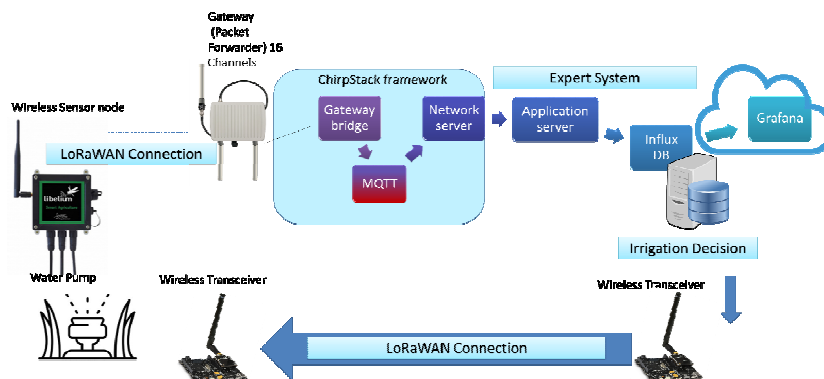


Fig. 1 The smart irrigation system network architecture

Within the atmosphere, the wind blows in an endeavor to equalize awkward nature in air pressure. Precisely, it blows within the heading from a region of high to low level of pressure at a velocity, which is relative to the pressure gradient [33, 34]. WDO algorithm is a new sort of nature-inspired optimization technique based on air motion. The WDO method could be a population based iterative aspect heuristic optimization algorithm for multi-dimensional issues with the capacity to actualize limitations on the search space. At its center, a population of imperceptibly small air parcels navigates over an n-dimensional search space taking after Newton's second law of movement, which is additionally utilized to depict the motion of air bundles inside the earth's climate. Compared to comparable particle-based algorithms, WDO utilizes extra terms within the velocity upgrade condition such as gravitation and Coriolis forces, giving strength and additional degrees of opportunity to fine-tune the optimization at the side the hypothesis and terminology of WDO [35].

Indeed, the major characteristics of the method is that WDO presents gravity and Coriolis forces within velocity modifying equation. While in other techniques, population applicants may jump out or be stuck at the boundary lines of each dimension, and that holds back the algorithm for discovering the whole search space efficiently. In WDO, if those who stay stuck, the gravitational force grabs the air parcels back to the search home. On another side, unlike other existing algorithms, a candidate of WDO is not influenced by the global best dimension, but each dimension is randomly affected by other Coriolis forces. Thus, provides more freedom in searching over a particular search space. WDO was used in several studies, such as electromagnetics optimization problems, and it showed superior performance in several cases compared with popular optimization techniques such as Particle Swarm Optimization (PSO) [36-38].

V. METHODOLOGY

This section explains the phases used in the proposed prediction model which are depicted in Figure 2 and explained briefly below.

A. Data Acquisition

Since the proposed irrigation system is still under development, the weather data of the capital of Jordan country (i.e. Amman) is acquired from meteoblue website, where data are freely available on the Internet [39]. In the near future, these data will be collected using the available sensor nodes. We collected 20 days of hourly data in the period between 1 until 20 November 2020. The collected weather data include the temperature, relative humidity, mean sea level pressure, and soil temperature, resulting with, 480 samples of data for each aforementioned measurements (i.e. 20 days * 24 hours).

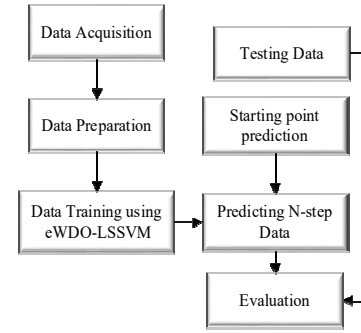


Fig. 2. The proposed prediction stages.

Therefore, the data were prepared to be fit correctly to the model. The data were separated into training and testing data, where the latter is set depending on the size of the sought forecasting period. Meanwhile, the remaining samples are set for training data, which therefore were windowed depending on the length of the window size of each prediction experiment; Thus, in this paper, we used the window sizes of 24, 48, and 72 for 24, 48, and 72 hours forecasting period.

B. WDO-LS-SVM Scheme

LS-SVM is a kernel-based machine learning algorithm provided as a solution for SVM regarding its equality limitation problem by solving a system of linear equations. This classifier has two parameters include smoothing parameter σ and RBF kernel parameter γ (i.e. regularization parameter). A different mixture of these values could provide different performance of the classifier. Consequently, this study proposed to use WDO algorithm to do this task, which results with the proposed new regression scheme named as WDO-LS-SVM. Therefore, for the implementation of the proposed scheme, the coefficients constant parameters of WDO are set as; gravitational constant $g=0.2$, alpha constant $\alpha=0.4$, coefficient $RT=2$, and Coriolis effect $c=0.4$. Moreover, the parameters include, population size, problem dimension, bands of search space, and iterations, were set to: 30, 2, (0.001-10000), and 100, respectively. The next section provides the experimental results.

Moreover, LS-SVM is used in the objective function where the training data is fed into the regressor (i.e. LS-SVM) for function estimation while the parameters are tuned by the WDO. The training data are trained by the regressor and then the predicted data of the resulted model and testing data are evaluated by calculating the Root Mean Square Error (RMSE) as a fitness value. The lowest RMSE indicates the best model of LS-SVM.

VI. EXPERIMENTAL RESULTS

In this study, six experiments were performed, mainly using two prediction models to be compared; (i) grid search-LS-SVM and (ii) WDO-LS-SVM. The algorithms were implemented, where the results of the two experiments in predicting 48 hours and 72 hours of weather data were recorded.

Figures 3 and 4 presenting line chart results of the four weather prediction factors for 48 and 72 hours, respectively. Generally, the graphs show that the forecasted values do not exactly match the actual values. Nevertheless, the lines represented WDO-LS-SVM data are nearer to the actual line in

comparison with the line data represented by the original LS-SVM. On another side, the superior performance of the WDO-LS-SVM can also be observed in several points on the charts. For example, it can be seen in Figure 3 at the pressure line of data; that how the predicted values actually mimic the actual values, unlike the line of data provided by the original LS-SVM. Moreover, there is generally a clear weakening of the performance of the original LS-SVM in comparison with the one obtained by the WDO-LS-SVM. Whereby, unlike the original LS-SVM that was tuned with grid search algorithm, the WDO-LS-SVM tries to adapt to the complexity changes of the regressor task (i.e. LS-SVM) depending on exploring more values that can ideally fit the regressor.

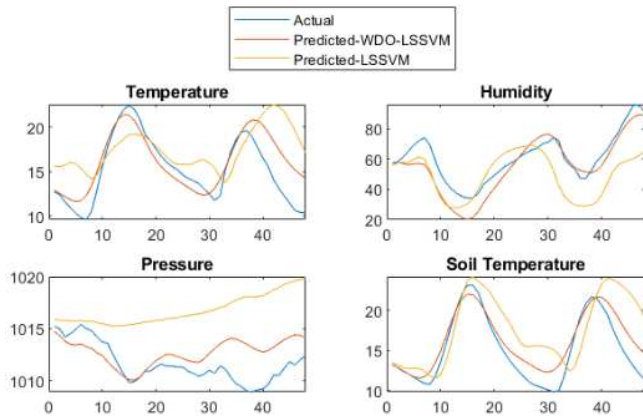


Fig. 3. Prediction for 48 hours.

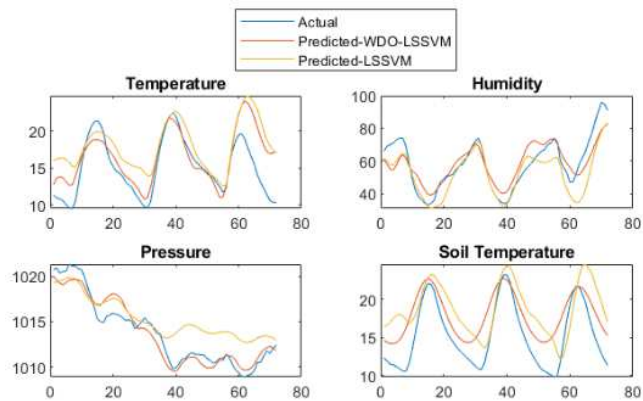


Fig. 4. Prediction for 72 hours.

Moreover, Table 1 illustrates the Normalized MSE (NRMSE) prediction results for the LS-SVM and WDO-LS-SVM models for data of the four weather factors. Generally, the NRMSE obtained by the proposed model is the lowest compared with the NRMSE values obtained by the original LS-SVM. For example, at the results of 72 hours prediction, the best results of NRMSE are obtained by WDO-LS-SVM, where the values are 0.2201, 0.1278, 0.0884, and 0.2367 for the temperature, relative humidity, mean sea level pressure, and soil temperature, respectively. Meanwhile, the best NRMSE obtained by the original LS-SVM were 0.3045, 0.1521, 0.1868, and 0.3182, respectively.

On another side, it can be observed that the results obtained by the LS-SVM fluctuate from one period to another, and/or they degrade. Unlike the results obtained by the WDO-LS-SVM, which are always straightforward towards searching optimal solution.

TABLE. I. THE NRMSE FOR TEMPERATURE AND HUMIDITY PREDICTION USING THE LSSVM AND WDO-LS-SVM

		Temperature	Humidity
48 hours	LSSVM	0.3332	0.2777
	WDO-LSSVM	0.168	0.1426
72 hours	LSSVM	0.3045	0.1521
	WDO-LSSVM	0.2201	0.1278

TABLE. II. THE NRMSE FOR THE FOUR PREDICTED WEATHER FACTORS USING THE LS-SVM AND WDO-LS-SVM

		Pressure	Soil temperature
48 hours	LS-SVM	0.8963	48 hours
	WDO-LS-SVM	0.3367	
72 hours	LSSVM	0.1868	72 hours
	WDO-LS-SVM	0.0884	

VII. CONCLUSION

This paper presents our initial system design for a smart irrigation system that utilizes LoRaWAN as the main wireless technology, and the ChirpStack framework for data processing, saving and presenting. In order to predict the optimal water irrigation needed, weather forecasting is often used as one of the input parameters for the irrigation prediction system. In this paper, a new prediction model to predict the aforementioned weather parameters that utilize the Wind Driven Optimization - Least Square Support Vector Machine algorithm is presented. The obtained results show that the proposed algorithm outperforms the Least Square Support Vector Machine algorithm that is widely used in weather prediction, which proves the effectiveness of the proposed algorithm.

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